

一、系统简介

此系统的核心功能为：位置定位、智能搜索、年龄适宜性判断与偏好记录与处理功能。

精准位置定位：利用高德地图的基于浏览器的精准定位 API，系统能够迅速获取用户当前位置信息。

智能地点搜索：在精准定位的基础上，利用高德地图搜索 POI 功能，结合用户输入的信息，系统能在海量数据中快速筛选出用户附近的地点信息。

适龄判断：考虑到不同场所可能存在的年龄限制（如酒吧），系统特别增设了年龄适宜性判断功能。系统会不执行未成年用户对附件酒吧的搜索。

用户偏好记录：系统会自动记录用户的 POI 搜索关键字。

用户偏好管理：管理用户偏好信息。

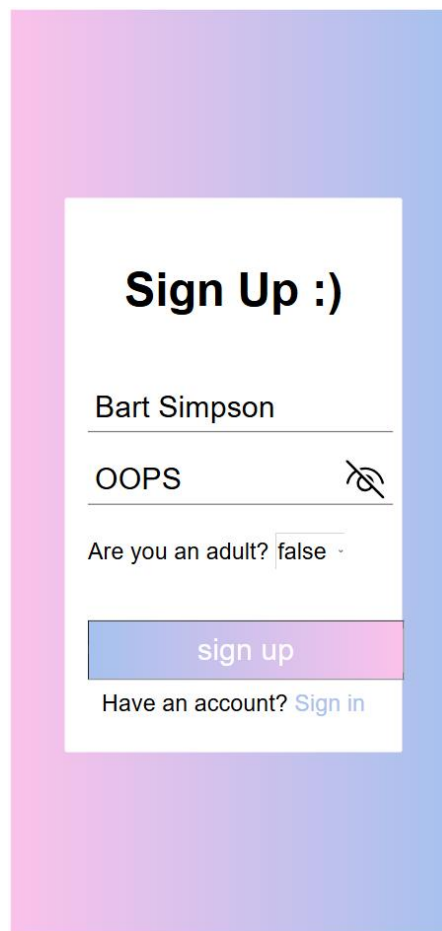
二、运行环境

此系统为前端为网页应用，支持在手机和电脑浏览器中访问。系统的后端数据库是 PostgreSQL。

三、主要功能及用法

使用案例：

3.1 用户注册及登录：



The image shows a mobile app interface for signing up. It features a white card on a pink and blue gradient background. The card has the title 'Sign Up :)' and two text input fields. The first field contains 'Bart Simpson' and the second contains 'OOPS'. To the right of the second field is a red 'X' icon. Below the fields is a dropdown menu labeled 'Are you an adult?' with 'false' selected. At the bottom of the card is a blue 'sign up' button and a link that says 'Have an account? Sign in'.

图表 1 注册界面



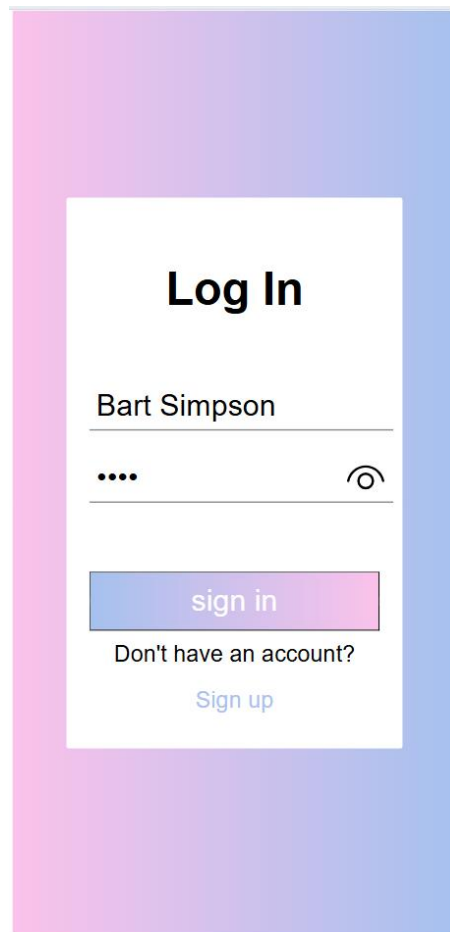
此页面显示

Sign up successful

图表 2 注册成功提示

	user_name [PK] character varying (128)	preference character varying (1024)	password character varying (128)	adult boolean
1	Sherlock Homles	[null]	1	true
2	John Watson	[null]	2	true
3	test	[null]	1234	false
4	Attie Lin	酒吧花猫	1	false
5	Bart Simpson	[null]	OOPS	false

图表 3 对应用户表（添加了 Bart Simpson 的记录）



图表 4 跳转到登录界面

此页面显示

Log in successful

图表 5 登录成功提示

对应代码
注册 html:



```
1. <!DOCTYPE html>
2. <html lang="en">
3. <head>
4.     <meta charset="UTF-8">
5.     <meta name="viewport2" content="width=device-width, initial-
        scale=1.0">
6.     <title>sign_up</title>
7.     <link rel = "stylesheet" href = "../login_style.css" type = "text/
        css">
8. </head>
9. <body>
10.     <div class="container">
11.         <div class="login-wrapper">
12.             <div class="header">Sign Up :)</div>
13. <!--
        -             <form action="http://127.0.0.1:5000/sign_up" method="pos
            t" class="form-wrapper">-->
14.                 <input type="text" id="name" placeholder="username" c
                    lass="input-item">
15.                 <input type="password" id="pwd" placeholder="password
                    " class="input-item">
16.                 <label class="msg" for="adult">Are you an adult? </la
                    bel>
17.                 <select name="adult" id="adult">
18.                     <option value="false" >false</option>
19.                     <option value="true">true</option>
20.                 </select>
21.                 <input class="btn" onclick="submit()" value="sign up"
                    ></input>
22.             </form>
23.             <div class="msg">
24.                 Have an account?
25.                 <a href="../login.html">Sign in</a>
26.             </div>
27.         </div>
28.     </div>
29. </div>
30.
31. <script src="../jquery.min.js"></script>
32. <script>
33.
34.     function submit() {
35.         name = document.getElementById("name").value
```



```
36.         pwd = document.getElementById("pwd").value
37.         adult = document.getElementById("adult").value
38. <!--         console.log(adult);-->
39.         if (name == "" || pwd == "")
40.         {
41.             alert("Please enter both password and name!");
42.         }
43.         $.ajax({
44.             type: "GET",
45.             url: "http://127.0.0.1:5000/sign_up",
46.             data: { "name": name, "pwd":pwd, "adult":adult},
47.             dataType: "json",
48.             success: function (res)
49.             {
50.                 if (res.msg == "error" )
51.                 {
52.                     window.location.href = "./sign_up.html";
53.                     alert('Sorry, but this name has already been
used,please think another one.');
```

登录 html

```
1. <!DOCTYPE html>
2. <html lang="en">
3. <head>
4.     <meta charset="UTF-8">
5.     <meta name="viewport2" content="width=device-width, initial-
scale=1.0">
6.     <title>Login</title>
7.     <link rel = "stylesheet" href = "./login_style.css" type = "text/
css">
```



```
8. </head>
9. <body>
10.     <div class="container">
11.         <div class="login-wrapper">
12.             <div class="header">Log In</div>
13. <!--
14.         <form action="http://127.0.0.1:5000/login" method="post"
15.             class="form-wrapper">-->
16.             <input type="text" id="name" placeholder="username" c
17.                 lass="input-item">
18.             <input type="password" id="pwd" placeholder="password
19.                 " class="input-item">
20.             <input type="button" class = 'btn' onclick="submit()"
21.                 value="sign in"></input>
22.         </form>-->
23.         <div class="msg">
24.             Don't have an account?
25.             <a href="./sign_up.html">Sign up</a>
26.         </div>
27.     </div>
28.
29.     <script src="./jquery.min.js"></script>
30.     <script>
31.         function submit()
32.         {
33.             name = document.getElementById("name").value;
34.             pwd = document.getElementById("pwd").value;
35.             if (name == "" || pwd == "")
36.             {
37.                 alert("Please enter both password and name!");
38.             }
39.             $.ajax({
40.                 type: "GET",
41.                 url: "http://127.0.0.1:5000/login",
42.                 data: { "name": name, "pwd":pwd},
43.                 dataType: "json",
44.                 success:function (res)
45.                 {
46.                     if (res.name == "None")
47.                     {
48.                         window.location.href = "./sign_up.html";
```



```
45.         alert("Sorry, but you should sign up first.")
46.     ;
47.     }
48.     else if (res.name != name || res.pwd != pwd)
49.     {
50.         window.location.href = "./login.html";
51.         alert("Wrong name or password");
52.     }
53.     else if (res.name == name && res.pwd == pwd)
54.     {
55.         window.location.href = "./get_location.html?"
56.         +"name="+name;
57.         alert("Log in successful");
58.     }
59.     else
60.     {
61.         window.location.href = "./login.html";
62.         alert("Unkown situation");
63.     }
64. }
65. });
66. </script>
67. </body>
68. </html>
```

注册接口代码:

```
1. @app.route("/sign_up", methods=['GET', 'POST'])
2. def sign_up():
3.     pwd = request.args.get("pwd")
4.     name = request.args.get("name")
5.     adult = request.args.get("adult")
6.     #search data in the customer table to see if the user is a kn
7.     own user.
8.     conn = psycopg2.connect(database=database, user=user, password=pa
9.     ssword, host=server, port=port) # 获取连接
10.    cursor = conn.cursor()
11.    search_user = "SELECT count(*) as num FROM users WHERE user_name=
12.    '{}'.format(name)
13.    cursor.execute(search_user)
14.    conn.commit()
15.    # fetch
16.    row = cursor.fetchone()
17.    # res = {"num": row[0]}
```



```
15.     cursor.close()
16.     # Check if the username is unique.
17.     if row[0] == 0:
18.         # This username is unique, so we can save it to our database.

19.         cursor = conn.cursor()
20.         ustr = "insert into users(user_name, password,adult) values('
                '{}','{}','{}')".format(name, pwd,adult)
21.         cursor.execute(ustr)
22.         res = {"msg": "success"}
23.     else:
24.         res = {"msg": "error"}
25.     conn.commit()
26.     cursor.close()
27.     return jsonify(res)
```

登录接口代码:

```
1. @app.route("/login", methods=['GET', 'POST'])
2. def login():
3.     pwd = request.args.get("pwd")
4.     name = request.args.get("name")
5.     #print(pwd,name)
6.     #search data in the customer table
7.     conn = psycopg2.connect(database=database, user=user, password=pa
        ssword, host=server, port=port) # 获取连接
8.     cursor = conn.cursor()
9.     search_user = "SELECT user_name,password FROM users WHERE user_na
        me='{}'".format(name)
10.    cursor.execute(search_user)
11.    conn.commit()
12.    # fetch
13.
14.    row = cursor.fetchone()
15.    if isinstance(row, tuple):
16.        res = {'name':row[0],'pwd':row[1] }
17.        # print("dfafdsafewg")
18.    else:
19.        res = {'name':"None",'pwd':"None" }
20.        # print("Dfafwe")
21.    cursor.close()
22.    conn.close()
23.    return jsonify(res)
```

3.2 用户定位及搜索 POI:



图表 6 用户登录好后，直接跳转到此界面



图表 8 搜索 POI



此页面显示

Sorry, but you aren't old enough to go to these kind of places.

确定



图表 9 此用户查询酒吧后的提示

对应获取位置的 html:

```
<!doctype html>
<html>
<head>
  <meta charset="utf-8">
  <meta http-equiv="X-UA-Compatible" content="IE=edge">
  <meta name="viewport" content="initial-scale=1.0, user-
scalable=no, width=device-width">
  <title>浏览器精确定位</title>
  <link rel="stylesheet"
href="https://a.amap.com/jsapi_demos/static/demo-center/css/demo-
center.css" />
  <style>
    html,body,#container{
      height:95.9%;
    }
    .info{
      width:26rem;
    }
    .btn {
      display:block;
      margin-left:auto;
      margin-right:auto;
      padding: 10px;
      width: 40%;
      margin-top: 40px;
      background-image: linear-gradient(to right, #a6clee,
#fbc2eb);
      color: #fff;
    }
  </style>
<body>
<div id='container'></div>
<div class="info">
  <h4 id='status'></h4><hr>
  <p id='result'></p><hr>
  <!--    <p >由于众多浏览器已不再支持非安全域的定位请求，为保位成功率和精度，请升
级您的站点到 HTTPS。</p>-->
</div>
<button class = 'btn' id = 'button'>Search nearby POIs.</button>
<script type="text/javascript"
src="https://webapi.amap.com/maps?v=1.4.15&key=eb5e1010e4dd136f251c52
9d530f23eb"></script>
<script type="text/javascript">
```



```
var name = location.search;
<!-- console.log(name);-->
name=name.slice(name.indexOf("=")+1);
<!-- console.log(name);-->
var lng, lat;
var map = new AMap.Map('container', {
    resizeEnable: true
});
AMap.plugin('AMap.Geolocation', function() {
    var geolocation = new AMap.Geolocation({
        enableHighAccuracy: true, //是否使用高精度定位, 默认:true
        timeout: 10000,           //超过 10 秒后停止定位, 默认: 5s
        buttonPosition: 'RB',     //定位按钮的停靠位置
        buttonOffset: new AMap.Pixel(10, 20), //定位按钮与设置的停靠位置
的偏移量, 默认: Pixel(10, 20)
        zoomToAccuracy: true,     //定位成功后是否自动调整地图视野到定位点
    });
    map.addControl(geolocation);
    geolocation.getCurrentPosition(function(status, result) {
        if(status=='complete'){
            onComplete(result)
        }else{
            onError(result)
        }
    });
});
//解析定位结果
function onComplete(data) {
    document.getElementById('status').innerHTML='定位成功'
    var str = [];
    str.push('定位结果: ' + data.position);
    str.push('定位类别: ' + data.location_type);
    if(data.accuracy){
        str.push('精度: ' + data.accuracy + ' 米');
    } //如为 IP 精确定位结果则没有精度信息
    str.push('是否经过偏移: ' + (data.isConverted ? '是' : '否'));
    document.getElementById('result').innerHTML =
str.join('<br>');

    lng = data.position.lng;
    lat = data.position.lat;
<!-- console.log(lng,lat);-->
}
//解析定位错误信息
```

```
function onError(data) {  
    document.getElementById('status').innerHTML='定位失败'  
    document.getElementById('result').innerHTML = '失败原因排查信  
息:' + data.message;  
}  
let button=document.getElementById("button")  
//send data to get_nearby_poi.html  
button.onclick=function () {  
    //路径+参数名+参数值  
    window.location.href =  
'./search_nearby.html?' + 'lng=' + lng + '&' + 'lat=' + lat + '&' + 'name=' + name;  
}  
</script>  
</body>  
</html>
```

对应查询 POI 的 html:

```
<!doctype html>  
<html>  
<head>  
    <meta charset="utf-8">  
    <meta http-equiv="X-UA-Compatible" content="IE=edge">  
    <meta name="viewport" content="initial-scale=1.0, user-  
scalable=no, width=device-width">  
    <title>search_nearby</title>  
    <link rel="stylesheet"  
href="https://cache.amap.com/lbs/static/main1119.css"/>  
    <style type="text/css">  
        #panel {  
            position: fixed;  
            background-color: white;  
            max-height: 90%;  
            overflow-y: auto;  
            top: 10px;  
            right: 10px;  
            width: 280px;  
            border-bottom: solid 1px silver;  
        }  
    </style>  
    <style>  
        .btn {  
            text-align: center;
```



```
padding: 2rem;
width: 60%;
height: 10%;
margin-top: 15rem;
background-image: linear-gradient(to right, #a6c1ee,
#fbc2eb);
color: #fff;
position: absolute; z-index: 10
}
.input-item {
display: block;
width: 50%;
margin-bottom: 20rem;
border: 0;
padding: 10px;
border-bottom: 1px solid rgb(128, 125, 125);
font-size: 15px;
outline: none;
position: absolute;
z-index: 10
}
</style>
<div id="container"></div>
<script type="text/javascript">
window._AMapSecurityConfig = {
securityJsCode: "f7bec2181f008568c61d9984750f1d35",
};</script>
<script
type="text/javascript"
src="https://webapi.amap.com/maps?v=2.0&key=1130a612091d60ccade5c924c35f2210"></script>

<script type="text/javascript"
src="https://webapi.amap.com/maps?v=2.0&key=1130a612091d60ccade5c924c35f2210"></script>
<script type="text/javascript"
src="https://cache.amap.com/lbs/static/addToolbar.js"></script>
</head>
<body>
<div id="container"></div>
<div id="panel"></div>
<input type="text" id="distance" placeholder="Maximum distance(Unit:
meter)" class="input-item" style = "margin-top: 31rem;">
```

```
<input type="text" id="keywords" placeholder="Keyword(s) of poi
(Please enter Chinese) "
    class="input-item" style = "margin-top: 36rem;">
<input type = "button" class = 'btn' id = 'button' style = "margin-
top: 40rem;"
    onclick="keywords_search()" value="Search nearby related
POIs."></input>
<script src="./jquery.min.js"></script>
<script type="text/javascript">
    var search_distance;
    var pos = location.search;
    console.log("0pos is ",pos);
    pos=pos.slice(pos.indexOf("=")+1)
    console.log("1pos is ",pos);
    var lng = pos.substring(0,10);
    var lat = pos.substring(15,24);
    var cpoint = [Number(lng), Number(lat)]; //中心点坐标

    var name = pos.substring(30);
    <!-- Turn all the space(%20) in the name to real space.-->
    name = name.replace("%20", " ");
    <!-- Turn all the /(%2F) in the name to real %2F.-->
    name = name.replace("%2F", "/");
    <!-- Turn all the +(%2B) in the name to real +.-->
    name = name.replace("%2B", "+");
    <!-- Turn all the ?(%3F) in the name to real ?.-->
    name = name.replace("%3F", "?");
    <!-- Turn all the %(%25) in the name to real %.-->
    name = name.replace("%25", "%");
    <!-- Turn all the #(%23) in the name to real #.-->
    name = name.replace("%23", "#");
    <!-- Turn all the &(%26) in the name to real &.-->
    name = name.replace("%26", "&");
    <!-- Turn all the =( %3D) in the name to real =.-->
    name = name.replace("%3D", "=");

    console.log("name is ",name);
    var keywords;
    var map = new AMap.Map("container", {
        resizeEnable: true
    });
    AMap.plugin("AMap.PlaceSearch", function()
    {
```



```
        autoFitView: true // 是否自动调整地图视野使绘制的
Marker 点都处于视口的可见范围
    });
    placeSearch.searchNearBy(keywords, cpoint,
Number(distance), function(status, result) {
    console.log(result);
    });
    });
    }
    }
    });
}

function keywords_search()
{
    distance = document.getElementById("distance").value
    keywords = document.getElementById("keywords").value
    detect_age_and_save_preference();
}
</script>
</body>
</html>
```

对应查询是否成年并存储偏好的 python 接口：

```
1. @app.route("/search_nearby", methods=['GET', 'POST']) # POST GET
2. def age_limit_and_save_preference():
3.     # 获取参数
4.     preference = request.args.get("preference")
5.     name = request.args.get("name")
6.
7.     #save preference
8.     conn = psycopg2.connect(database = database, user = user, password = password, host = server,port = port) #获取连接
9.     cursor_fetch = conn.cursor()
10.
11.     sql_fetch = "SELECT preference FROM users WHERE user_name='{ }'".format(name)
12.     cursor_fetch.execute(sql_fetch)
13.     # fetch
14.     the_user_preference = cursor_fetch.fetchone()
15.     # res = {"adult":row[0]}
```



```
16.     cursor_update = conn.cursor()
17.     if the_user_preference[0] != None:
18.         sql_update = "update users set preference = concat('{}','{}')
        WHERE user_name='{}'".format(preference,
19.
                                   the_user_preference[0],
20.
                                   name)
21.     else:
22.         preference = ";" + preference
23.         sql_update = "update users set preference = concat('{}') WHER
        E user_name='{}'".format(preference,
24.
                                   name)
25.     cursor_fetch.execute(sql_update)
26.     cursor_update.close()
27.     cursor_fetch.close()
28.
29.     #detect if the user is an adult.
30.     cursor = conn.cursor()
31.     sql = "SELECT adult FROM users WHERE user_name='{}'".format(name)
32.
33.     # sql = "SELECT adult FROM users WHERE user_name='Attie Lin'"
34.     cursor.execute(sql)
35.     conn.commit()
36.     #fetch
37.     row = cursor.fetchone()
38.     res = {"adult":row[0]}
39.     cursor.close()
40.     conn.close()
41.     return jsonify(res)
```

3.3 用户偏好信息处理

已修改用户“Attie Lin”的信息为例：

user_name [PK] character varying (128)	preference character varying (1024)	password character varying (128)	adult boolean
Attie Lin	酒吧花猫***@#星空	1	false

图表 10 修改前

Edit Preference

Attie Lin

Edit this record

图表 11 修改界面

此页面显示

Change successful :)

图表 12 修改成功提示

	user_name [PK] character varying (128)	preference character varying (1024)	password character varying (128)	adult boolean
1	Sherlock Homles	[null]	1	true
2	John Watson	[null]	2	true
3	test	[null]	1234	false
4	Bart Simpson	酒吧吃	OOPS	false
5	Attie Lin	酒吧花猫星空	1	false

图表 13 修改后的表

修改用户偏好信息的 python 代码:

```

1. def listToString(s):
2.     # initialize an empty string
3.     str1 = ""
4.     # traverse in the string
5.     for ele in s:
6.         str1 += ele
7.     # return string
8.     return str1

1. @app.route("/edit_preference", methods=['GET', 'POST'])
2. def edit_preference():
3.     with open('stop_words.txt', encoding='utf-8') as f:
4.         con = f.readlines()
5.         stop_words = set() # 集合可以去重

```



```
6.         for i in con:
7.             i = i.replace("\n", "") # 去掉读取每一行数据的\n
8.             stop_words.add(i)
9.
10.        name = request.args.get("name")
11.        # fetch the user's preference data.
12.        conn = psycopg2.connect(database=database, user=user, password=password, host=server, port=port) # 获取连接
13.        cursor = conn.cursor()
14.        cursor_update = conn.cursor()
15.        search_user = "SELECT preference FROM users WHERE user_name='{ }'".format(name)
16.        # search_user = "SELECT preference FROM users WHERE user_name='Attie Lin'"
17.        cursor.execute(search_user)
18.        conn.commit()
19.        # fetch
20.        row = cursor.fetchone()
21.        # res = {"num": row[0]}
22.        preference = row[0]
23.
24.        result = []
25.        for word in jieba.lcut(preference):
26.            if word not in stop_words:
27.                result.append(word)
28.        result = listToString(result)
29.
30.        #update the result into the user's preference column.
31.        sql_update = "update users set preference = '{ }' WHERE user_name='{ }'".format(result, name)
32.        # sql_update = "UPDATE users SET preference = '{ }' WHERE user_name = 'Attie Lin'".format(result)
33.        cursor_update.execute(sql_update)
34.        conn.commit()
35.        print(result)
36.        cursor.close()
37.        cursor_update.close()
38.        conn.close()
39.
40.        res = {"result": "success"}
41.        return jsonify(res)
```